

Supplementary Information (MLShrink)

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ABSTRACT

1. Algorithmic Representation

Algorithm 1 Algorithm 1 *MLShrink* via Classification on the Undecided Band

Require: Wavelet coefficients $d_{j,k}$, thresholds $\lambda_1 < \lambda_2$, classifier C

Ensure: Denoised wavelet coefficients

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1: Initialize lists:  $\mathbf{X} \leftarrow [], \mathbf{Y} \leftarrow [], \mathcal{I}_{\text{undecided}} \leftarrow []$ 
2: for each scale  $j$  and location  $k$  do
3:    $v_{j,k} \leftarrow (|d_{j,k-1}| + |d_{j,k+1}|)/2$ 
4:    $x_{j,k} \leftarrow [|d_{j,k}|, j, v_{j,k}]^T$ 
5:   Append  $\mathbf{x}_{j,k}$  to  $\mathbf{X}$ 
6:   if  $|d_{j,k}| \leq \lambda_1$  then
7:      $L_{j,k} \leftarrow 0$ 
8:   else if  $|d_{j,k}| \geq \lambda_2$  then
9:      $L_{j,k} \leftarrow 1$ 
10:  else
11:     $L_{j,k} \leftarrow u$ 
12:    Append  $(j, k)$  to  $\mathcal{I}_{\text{undecided}}$ 
13:  end if
14:  Append  $L_{j,k}$  to  $\mathbf{Y}$ 
15: end for
16:  $\mathbf{X}_{\text{train}} \leftarrow \{\mathbf{x}_{j,k} \in \mathbf{X} : L_{j,k} \in \{0, 1\}\}$ 
17:  $\mathbf{Y}_{\text{train}} \leftarrow \{L_{j,k} \in \mathbf{Y} : L_{j,k} \in \{0, 1\}\}$ 
18:  $\mathbf{X}_{\text{pred}} \leftarrow \{\mathbf{x}_{j,k} \in \mathbf{X} : L_{j,k} = u\}$ 
19:  $C.\text{fit}(\mathbf{X}_{\text{train}}, \mathbf{Y}_{\text{train}})$ 
20:  $\hat{\mathbf{Y}}_{\text{pred}} \leftarrow C.\text{predict}(\mathbf{X}_{\text{pred}})$ 
21: for each  $(j, k) \in \mathcal{I}_{\text{undecided}}$  with predicted label  $\hat{L}_{j,k}$  do
22:   if  $\hat{L}_{j,k} = 0$  then
23:      $d_{j,k} \leftarrow 0$ 
24:   end if
25: end for
26: return denoised coefficients
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2. Additional theoretical refinements

The main text establishes the structural properties of *MLShrink*, an oracle-based risk decomposition, an oracle-consistency result, a Bayesian interpretation under a contamination prior, and a theorem giving sufficient conditions under which *MLShrink* improves upon hard thresholding. The next two results are more technical refinements and

are included in the appendix to keep the main exposition focused.

The first result sharpens the oracle-consistency perspective by relating excess denoising risk to the quality of an estimated oracle score on the undecided band. Its role is to show that the performance of *MLShrink* depends not only on whether classification error vanishes, but also on how well the learned score approximates the oracle posterior decision boundary. Under a standard margin condition, this yields a quantitative excess-risk bound.

The second result gives a levelwise version of the oracle comparison. Since wavelet methods are intrinsically multiscale, it is natural to understand how the gap between *MLShrink* and its oracle benchmark is distributed across levels. This theorem makes precise that the excess risk can be decomposed level by level, which may be useful in future extensions with level-dependent thresholds or level-specific classifiers.

Theorem 2.1 (Margin-based excess-risk bound on the undecided band). *For $(j, k) \in R_u$, let*

$$\eta_{j,k} = P(L_{j,k}^* = 1 \mid x_{j,k}, (j, k) \in R_u) \quad (1)$$

denote the oracle posterior score, and let the classifier produce a score $\hat{\eta}_{j,k} \in [0, 1]$ with plug-in label

$$\hat{L}_{j,k} = \mathbf{1}\{\hat{\eta}_{j,k} \geq 1/2\}. \quad (2)$$

Assume that there exist constants $A > 0$ and $\kappa > 0$ such that for all $t > 0$,

$$P(|\eta_{j,k} - 1/2| \leq t, (j, k) \in R_u) \leq At^\kappa. \quad (3)$$

Assume also that $|d_{j,k}| \leq \lambda_2$ and $|\theta_{j,k}| \leq M$ on R_u . Define

$$\delta_{j,k} = E(|\hat{\eta}_{j,k} - \eta_{j,k}| \mathbf{1}\{(j, k) \in R_u\}). \quad (4)$$

Then there exists a constant $C = C(A, \kappa, M, \lambda_2)$ such that

$$E\|\hat{d}^{MLShrink} - \theta\|^2 - E\|\hat{d}^{oracle} - \theta\|^2 \leq C \sum_{j,k} \delta_{j,k}^{\frac{\kappa}{\kappa+1}}. \quad (5)$$

In particular, if

$$\sup_{j,k} \delta_{j,k} \leq \delta_n, \quad (6)$$

then

$$E\|\hat{d}^{MLShrink} - \theta\|^2 - E\|\hat{d}^{oracle} - \theta\|^2 = O(|R_u| \delta_n^{\frac{\kappa}{\kappa+1}}). \quad (7)$$

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Proof. By the exact oracle-risk decomposition in Theorem 1,

$$\begin{aligned} & E \left\| \hat{d}^{MLShrink} - \theta \right\|^2 - E \left\| \hat{d}^{oracle} - \theta \right\|^2 \\ &= \sum_{j,k} E \left[\mathbf{1}\{(j,k) \in R_u\} \Delta_{j,k} \mathbf{1}\{\hat{L}_{j,k} \neq L_{j,k}^*\} \right]. \end{aligned} \quad (8)$$

Now, on R_u we have $|d_{j,k}| < \lambda_2$, and by assumption $|\theta_{j,k}| \leq M$. Hence, using the same argument as in Corollary 1,

$$\begin{aligned} \Delta_{j,k} &= \left| 2d_{j,k}\theta_{j,k} - d_{j,k}^2 \right| \leq 2|d_{j,k}||\theta_{j,k}| + |d_{j,k}|^2 \leq 2\lambda_2 M \\ &\quad + \lambda_2^2. \end{aligned} \quad (9)$$

Set

$$B = \lambda_2(2M + \lambda_2). \quad (10)$$

Then (8) implies

$$\begin{aligned} & E \left\| \hat{d}^{MLShrink} - \theta \right\|^2 - E \left\| \hat{d}^{oracle} - \theta \right\|^2 \\ &\leq B \sum_{j,k} P(\hat{L}_{j,k} \neq L_{j,k}^*, (j,k) \in R_u). \end{aligned} \quad (11)$$

Fix (j,k) . We now bound the disagreement probability. For any $t > 0$, consider the event

$$E_{j,k} = \left\{ \hat{L}_{j,k} \neq L_{j,k}^*, (j,k) \in R_u \right\}. \quad (12)$$

We claim that

$$\begin{aligned} E_{j,k} &\subseteq \left\{ |\eta_{j,k} - 1/2| \leq t, (j,k) \in R_u \right\} \\ &\cup \left\{ |\hat{\eta}_{j,k} - \eta_{j,k}| > t, (j,k) \in R_u \right\}. \end{aligned} \quad (13)$$

Indeed, suppose that both $|\eta_{j,k} - 1/2| > t$ and $|\hat{\eta}_{j,k} - \eta_{j,k}| \leq t$ hold. Then $\hat{\eta}_{j,k}$ lies on the same side of $1/2$ as $\eta_{j,k}$, so the plug-in label $\mathbf{1}\{\hat{\eta}_{j,k} \geq 1/2\}$ must agree with the Bayes label $\mathbf{1}\{\eta_{j,k} \geq 1/2\}$. Hence disagreement is impossible, proving (13).

Therefore,

$$\begin{aligned} P(E_{j,k}) &\leq P(|\eta_{j,k} - 1/2| \leq t, (j,k) \in R_u) + \\ &\quad P(|\hat{\eta}_{j,k} - \eta_{j,k}| > t, (j,k) \in R_u) \\ &\leq At^\kappa + P(|\hat{\eta}_{j,k} - \eta_{j,k}| \mathbf{1}\{(j,k) \in R_u\} > t), \end{aligned} \quad (14)$$

where the second line uses the margin assumption (3).

Now apply Markov's inequality to the second term:

$$\begin{aligned} & P(|\hat{\eta}_{j,k} - \eta_{j,k}| \mathbf{1}\{(j,k) \in R_u\} > t) \\ &\leq \frac{E(|\hat{\eta}_{j,k} - \eta_{j,k}| \mathbf{1}\{(j,k) \in R_u\})}{t} = \frac{\delta_{j,k}}{t}. \end{aligned} \quad (15)$$

Combining (14) and (15), we obtain

$$P(E_{j,k}) \leq At^\kappa + \frac{\delta_{j,k}}{t}, \quad t > 0. \quad (16)$$

We now optimize the right-hand side in t . Let

$$t_\star = \left(\frac{\delta_{j,k}}{A\kappa} \right)^{\frac{1}{\kappa+1}}. \quad (17)$$

Substituting $t_\star$ into (16) yields

$$\begin{aligned} P(E_{j,k}) &\leq A^{\frac{1}{\kappa+1}} \delta_{j,k}^{\frac{\kappa}{\kappa+1}} \left(\kappa^{-\frac{\kappa}{\kappa+1}} + \kappa^{\frac{1}{\kappa+1}} \right) \\ &\leq C_0 \delta_{j,k}^{\frac{\kappa}{\kappa+1}}, \end{aligned} \quad (18)$$

where

$$C_0 = A^{\frac{1}{\kappa+1}} \left(\kappa^{-\frac{\kappa}{\kappa+1}} + \kappa^{\frac{1}{\kappa+1}} \right). \quad (19)$$

Finally, substitute (18) into (11):

$$\begin{aligned} & E \left\| \hat{d}^{MLShrink} - \theta \right\|^2 - E \left\| \hat{d}^{oracle} - \theta \right\|^2 \\ &\leq BC_0 \sum_{j,k} \delta_{j,k}^{\frac{\kappa}{\kappa+1}}. \end{aligned} \quad (20)$$

This proves (5), with

$$C = BC_0. \quad (21)$$

If $\sup_{j,k} \delta_{j,k} \leq \delta_n$, then

$$\sum_{j,k} \delta_{j,k}^{\frac{\kappa}{\kappa+1}} \leq |R_u| \delta_n^{\frac{\kappa}{\kappa+1}}, \quad (22)$$

and (7) follows. $\square$

Theorem 2.2 (Levelwise oracle inequality). *For each resolution level j , define*

$$\begin{aligned} R_j^{MLShrink} &= \sum_k E \left(\hat{d}_{j,k}^{MLShrink} - \theta_{j,k} \right)^2, \\ R_j^{oracle} &= \sum_k E \left(\hat{d}_{j,k}^{oracle} - \theta_{j,k} \right)^2. \end{aligned} \quad (23)$$

Then

$$\begin{aligned} & R_j^{MLShrink} - R_j^{oracle} \\ &= \sum_k E \left[\mathbf{1}\{(j,k) \in R_u\} \Delta_{j,k} \mathbf{1}\{\hat{L}_{j,k} \neq L_{j,k}^*\} \right]. \end{aligned} \quad (24)$$

If moreover $|\theta_{j,k}| \leq M_j$ on level j and $|d_{j,k}| \leq \lambda_2$ on R_u , then

$$\begin{aligned} & R_j^{MLShrink} - R_j^{oracle} \\ &\leq \lambda_2(2M_j + \lambda_2) \sum_k P(\hat{L}_{j,k} \neq L_{j,k}^*, (j,k) \in R_u). \end{aligned} \quad (25)$$

Summing over j recovers the global oracle bound.

Proof. For a fixed level j , the coefficientwise identity used in the proof of Theorem 1 gives

$$\begin{aligned} \left(\hat{d}_{j,k}^{MLShrink} - \theta_{j,k} \right)^2 &= \left(\hat{d}_{j,k}^{oracle} - \theta_{j,k} \right)^2 \\ &+ \mathbf{1}\{(j, k) \in R_u\} \Delta_{j,k} \mathbf{1}\{\hat{L}_{j,k} \neq L_{j,k}^*\}. \end{aligned} \quad (26)$$

Summing (26) over k and then taking expectations yields

$$\begin{aligned} R_j^{MLShrink} - R_j^{oracle} \\ = \sum_k E \left[\mathbf{1}\{(j, k) \in R_u\} \Delta_{j,k} \mathbf{1}\{\hat{L}_{j,k} \neq L_{j,k}^*\} \right], \end{aligned} \quad (27)$$

which proves (24).

Now assume that $|\theta_{j,k}| \leq M_j$ on level j and $|d_{j,k}| \leq \lambda_2$ on R_u . Then, for $(j, k) \in R_u$,

$$\begin{aligned} \Delta_{j,k} &= \left| 2d_{j,k}\theta_{j,k} - d_{j,k}^2 \right| \leq 2|d_{j,k}||\theta_{j,k}| + |d_{j,k}|^2 \\ &\leq 2\lambda_2 M_j + \lambda_2^2 = \lambda_2(2M_j + \lambda_2). \end{aligned} \quad (28)$$

Substituting this into (24), we obtain

$$\begin{aligned} R_j^{MLShrink} - R_j^{oracle} \\ \leq \lambda_2(2M_j + \lambda_2) \sum_k E \left[\mathbf{1}\{(j, k) \in R_u\} \mathbf{1}\{\hat{L}_{j,k} \neq L_{j,k}^*\} \right] \\ = \lambda_2(2M_j + \lambda_2) \sum_k P \left(\hat{L}_{j,k} \neq L_{j,k}^*, (j, k) \in R_u \right), \end{aligned} \quad (29)$$

which proves (25).

Finally, summing (24) or (25) over j recovers the corresponding global oracle decomposition or global oracle bound. $\square$

3. Classifier Hyperparameters

The hyperparameters of the random forest classifier used to produce Figures 3 and 4 in the main document are given in Tables 1 and 2, respectively. Table 2 summarizes classifier hyperparameters used to obtain Table 1 in the main document.

Table 1

Hyperparameter configuration of the Random Forest classifier (Figure 3)

Hyperparameter	Value / Setting
λ_1	2.8841
λ_2	3.7233
Number of Trees	150
Maximum Splits	20 (Bagging)
Minimum Leaf Size	10
Learning Rate	0.05
Cross-Validation	5-fold

Table 2

Optimized parameters of the Random Forest (Bagging) model used in Figure 4 for the Block signal under different SNR levels.

Hyperparameter	SNR = 3	SNR = 5	SNR = 7
Optimal c value	1.20	1.40	1.20
λ_1	2.8841	3.097	2.8841
λ_2	3.7233	3.7233	3.7233
Number of Trees	150	150	150
Maximum Splits	20	20	20
Minimum Leaf Size	10	10	10
Learning Rate	0.05	0.05	0.05
Cross-Validation	5-fold	5-fold	5-fold

Table 3

Classifier hyperparameters (LR- Logistic Regression, SVM- Support Vector Machine, RF- Random Forest, DT - Decision Tree, NN - Neural Network).

Classifier	Parameter	Value / Setting
LR	Regularization type	L2 (ridge)
	Solver	fitclinear
	Maximum iterations	1000
SVM	Kernel function	RBF
	Box constraint	1
	Cross validation	10-fold
RF	Number of Trees	150
	Maximum Splits	20 (Bagging)
	Minimum Leaf Size	10
	Learning Rate	0.05
	Cross-Validation	5-fold
DT	Maximum Decision splits	range [1, 100]
	Minimum leaf size	range [1, 50]
	Cross-validation folds	5 fold
NN	Hidden layer structure	[10, 5]
	Activation function	Sigmoid
	Training function	trainscg
	Epochs	1000

4. Performance comparison with existing shrinkage methods

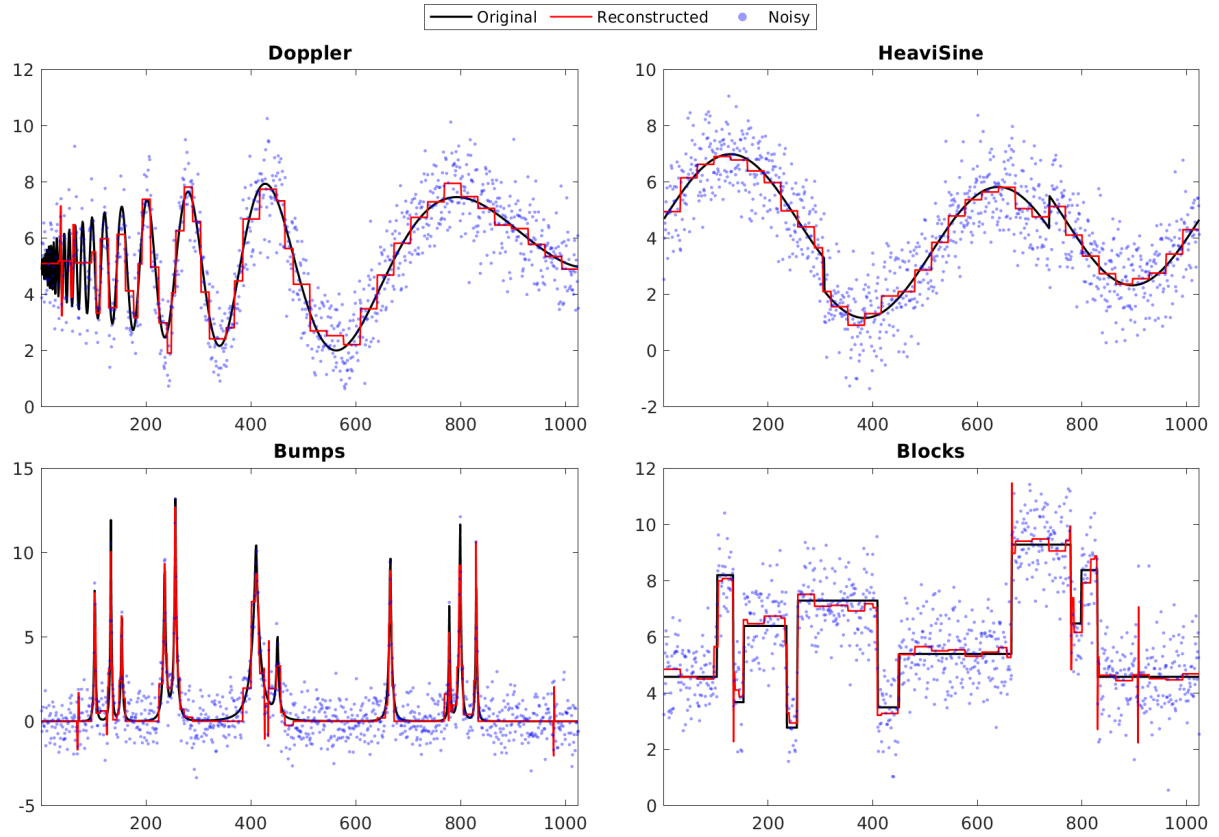


Figure 1: Reconstruction of the four standard test signals under $\text{SNR} = 3$.

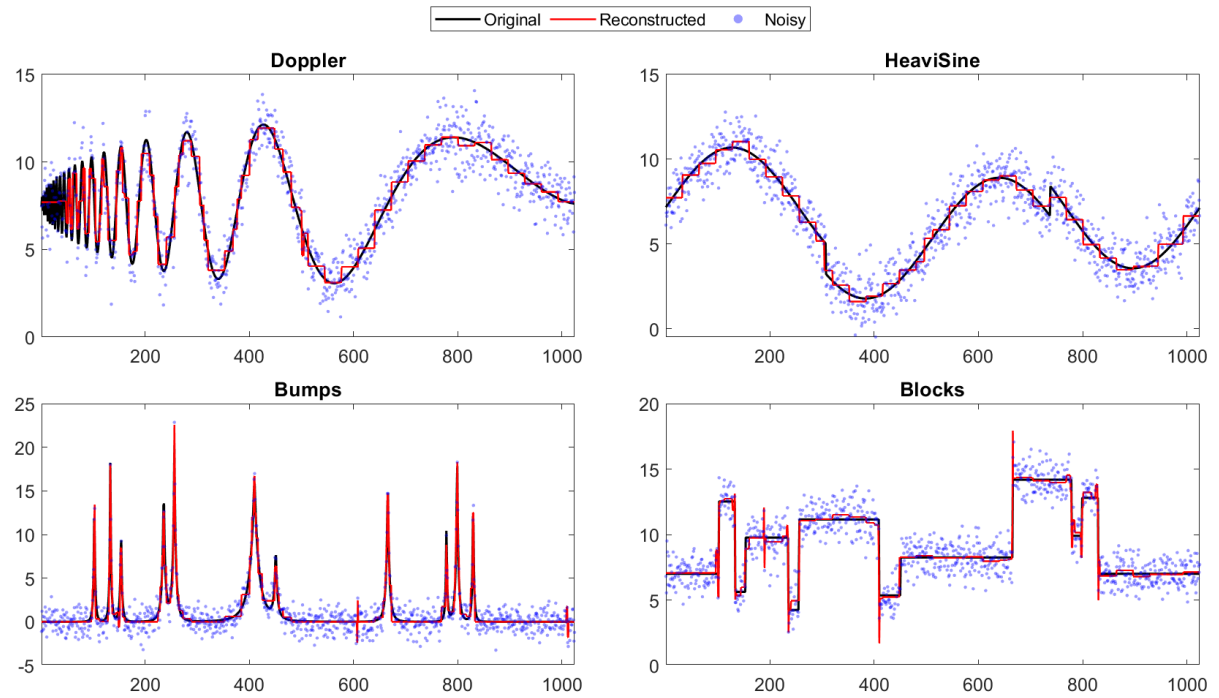


Figure 2: Reconstruction of the four standard test signals under $\text{SNR} = 7$.

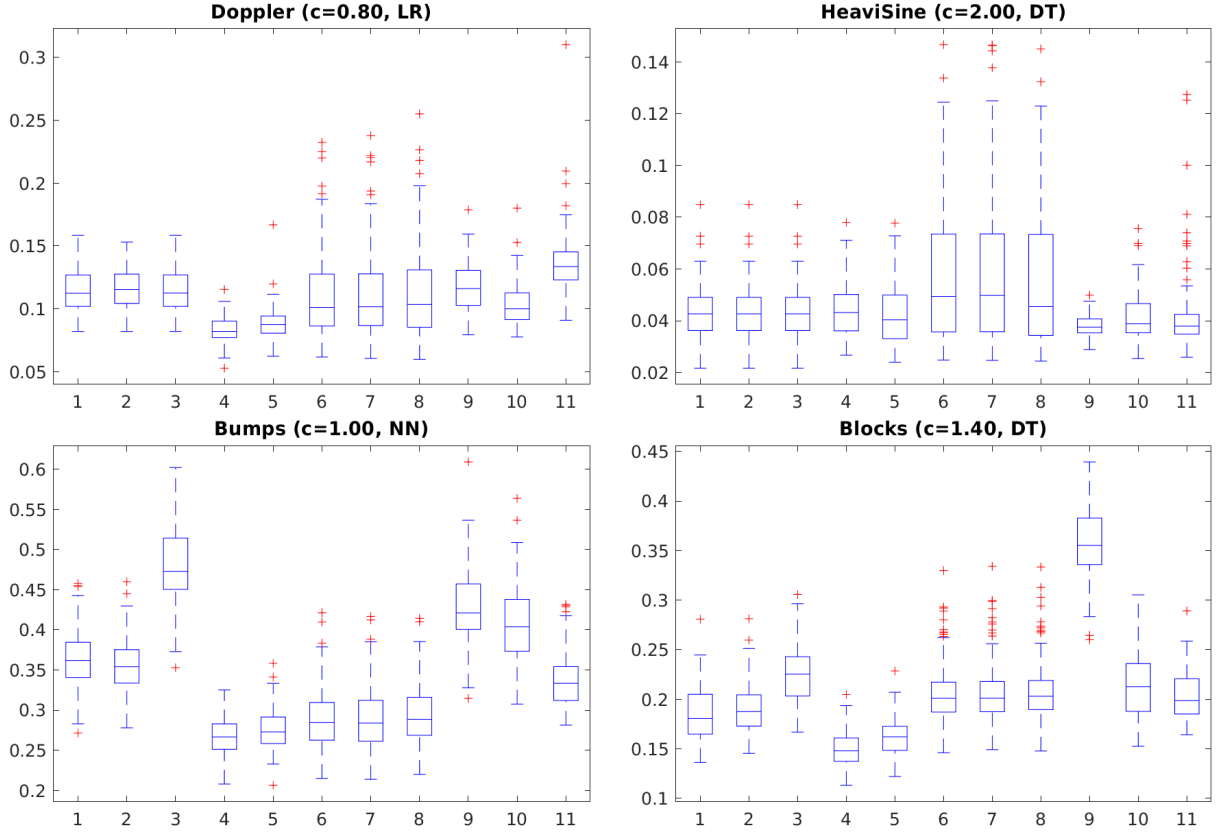


Figure 3: Boxplots of AMSE for *MLShrink* and competing denoising procedures at SNR = 3. The methods shown are: (1) *MLShrink*, (2) Semi-soft, (3) Hard, (4) Bayesian adaptive multiresolution shrinker (BAMS), (5) DCOMPSh, (6) Block-median, (7) Block-mean, (8) Hybrid block-median, (9) BlockJS, (10) VisuShrink, and (11) Generalized cross-validation.

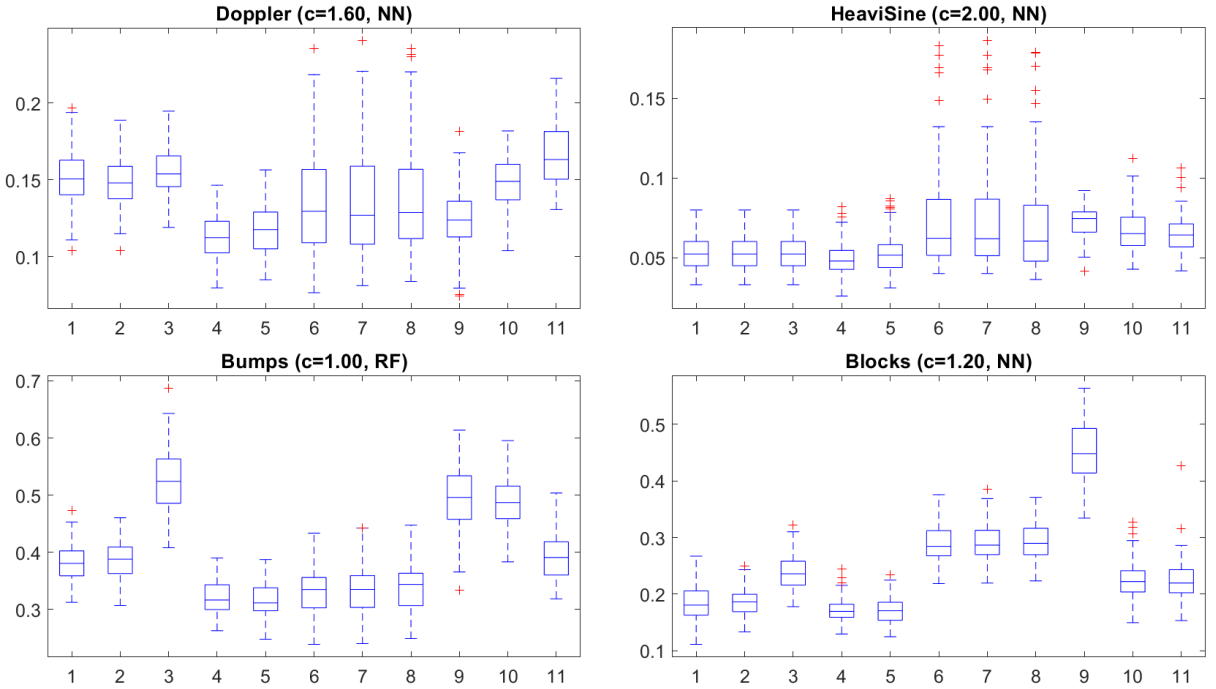


Figure 4: Boxplots of AMSE for *MLShrink* and competing denoising procedures at SNR = 7. The methods shown are: (1) *MLShrink*, (2) Semi-soft, (3) Hard, (4) Bayesian adaptive multiresolution shrinker (BAMS), (5) DCOMPSh, (6) Block-median, (7) Block-mean, (8) Hybrid block-median, (9) BlockJS, (10) VisuShrink, and (11) Generalized cross-validation.